Digital Signature-ElGamal Digital Signature Scheme

ElGamal Digital Signature Scheme:

ElGamal encryption scheme is designed to enable encryption by a user's public key with decryption by the user's private key. The ElGamal signature scheme involves the use of the private key for digital signature generation and the public key for digital signature verification.

Preliminary:

If q is a prime number and α is a primitive root of q, then

Elgamal Key Generation:

As with Elgamal encryption, the global elements of **Elgamal digital signature** are a prime number $\boldsymbol{q}$ and $\boldsymbol{\alpha}$, which is a primitive root of $\boldsymbol{q}$. User **A** generates a private/public key pair as follows:

1. Generate a random integer X_A, such that $1 < X_A < q - 1$.
2. Compute $Y_A = \alpha^{X_A} \bmod q$.
3. A's private key is X_A; A's pubic key is $\{q, \alpha, Y_A\}$.

Elgamal Digital Signature Generation:

To sign a message $\boldsymbol{M}$, user **A** first computes the hash $\boldsymbol{m = H(M)}$, such that $\boldsymbol{m}$ is an integer in the range $\boldsymbol{0 \leq m \leq q - 1}$. **A** then forms a digital signature as follows:

ElGamal Digital Signature Scheme:

Elgamal Digital Signature Generation (Continued):

1. Choose a random integer K such that $1 \leq K \leq q - 1$ and $\gcd(K, q - 1) = 1$. That is, K is relatively prime to $q - 1$.
2. Compute $S_1 = \alpha^K \bmod q$. Note that this is the same as the computation of C_1 for Elgamal encryption.
3. Compute $K^{-1} \bmod (q - 1)$. That is, compute the inverse of K modulo $q - 1$.
4. Compute $S_2 = K^{-1}(m - X_A S_1) \bmod (q - 1)$.
5. The signature consists of the pair (S_1, S_2).

Elgamal Digital Signature Verification:

Any user **B** can verify the signature as follows:

1. Compute $V_1 = \alpha^m \bmod q$.
2. Compute $V_2 = (Y_A)^{S_1}(S_1)^{S_2} \bmod q$.

The signature is valid if $\boldsymbol{V_1} = \boldsymbol{V_2}$. Let us demonstrate that this is so.

Example (1):

Let us start with the prime field GF(19); that is, $\boldsymbol{q}$ = 19. It has primitive roots {2, 3, 10, 13, 14, 15}. We choose $\boldsymbol{\alpha}$ = 10.

Alice generates a key pair as follows:

1. Alice chooses $X_A = 16$.
2. Then $Y_A = \alpha^{X_A} \bmod q = \alpha^{16} \bmod 19 = 4$.
3. Alice's private key is 16; Alice's pubic key is $\{q, \alpha, Y_A\} = \{19, 10, 4\}$.

ElGamal Digital Signature Scheme:

Example (1): (Continued)

(Sign) Suppose Alice wants to sign a message with hash value ***m*** = 14.

1. Alice chooses $K = 5$, which is relatively prime to $q - 1 = 18$.
2. $S_1 = \alpha^K \bmod q = 10^5 \bmod 19 = 3$
3. $K^{-1} \bmod (q - 1) = 5^{-1} \bmod 18 = 11$.
4. $S_2 = K^{-1}(m - X_A S_1) \bmod (q - 1) = 11\,(14 - (16)(3)) \bmod 18 = -374 \bmod 18 = 4$.

(Verify): Bob can verify the signature as follows:

1. $V_1 = \alpha^m \bmod q = 10^{14} \bmod 19 = 16$.
2. $V_2 = (Y_A)^{S_1}(S_1)^{S_2} \bmod q = (4^3)(3^4) \bmod 19 = 5184 \bmod 19 = 16$.

Thus, the signature is valid because $\boldsymbol{V_1} = \boldsymbol{V_2}$.

NIST Digital Signature Approach:

NIST has published Federal Information Processing Standard **FIPS 186**, known as the Digital Signature Algorithm (**DSA**). The **DSA** makes use of the Secure Hash Algorithm (**SHA**). The **DSA** was originally proposed in 1991. Several expanded versions of the standard were then issued as FIPS 186-2, FIPS 186-3 and FIPS 186-4 in response to public feedback concerning the security of the scheme. This latest version also incorporates digital signature algorithms based on RSA and on elliptic curve cryptography.